

TWIST-COMPATIBLE BURAU QUOTIENTS OF BIGELOW'S ZIPPER ALGEBRA

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ABSTRACT. Stephen Bigelow defined a braid-group-algebra quotient Z_n and asked whether its recursively defined element X_3 vanishes. A recent question of Bigelow asks whether a nonvanishing representation can also satisfy twist eigenvalue relations for X_2 and X_3 , with $X_4 = 0$. We give an exact affirmative answer at the level of representations. For the unreduced Burau representation with parameter u we prove

$$\rho(X_k) = \left(\prod_{j=1}^{k-1} (q^j - u) \right) E_{k-1}, \quad \text{rank } E_{k-1} = 1,$$

and the twist formulas

$$\rho(\sigma_1 X_2) = -u \rho(X_2), \quad \rho((\sigma_1 \sigma_2)^3 X_3) = u^3 \rho(X_3).$$

More generally, the positive full twist on the first k strands acts on $\rho(X_k)$ by u^k . Specializing $u = q^3$ gives simultaneously

$$\sigma_1 X_2 = -q^3 X_2, \quad (\sigma_1 \sigma_2)^3 X_3 = q^9 X_3, \quad X_4 = 0$$

in the Burau image, while X_3 remains nonzero. Consequently the quotient defined by these three additional relations is strictly above the specialized BMW quotient. Taking the joint image of its Burau and BMW representations produces a finite-dimensional braid-group algebra which surjects onto BMW with nonzero kernel. We do not claim that the universal quotient by the three relations is itself finite-dimensional.

1. INTRODUCTION

Let B_n be the braid group with standard generators $\sigma_1, \dots, \sigma_{n-1}$. In the final section of [1], Bigelow introduced a quotient Z_n of the braid group algebra and elements $X_2, X_3, \dots$ intended to interpolate between Hecke, BMW, and larger braid-group algebras. The printed presentation asks whether $X_3 = 0$ and which additional relations make the algebra finite-dimensional. It also observes that imposing

$$X_3 = 0, \quad \sigma_1 X_2 = t X_2$$

gives the Birman–Wenzl–Murakami algebra.

The question addressed here arose from a subsequent communication of Bigelow: can a representation with $X_3 \neq 0$ also make X_2 and X_3 eigenvectors for their natural twists, and can one additionally impose $X_4 = 0$? The answer is yes. The scalars are not independent in the Burau branch: if

$$\sigma_1 X_2 = \alpha X_2, \quad (\sigma_1 \sigma_2)^3 X_3 = \beta X_3,$$

then our construction has $\alpha = -u$ and $\beta = u^3 = -\alpha^3$. The condition $X_4 = 0$ selects $u = q^3$, and hence

$$(\alpha, \beta) = (-q^3, q^9).$$

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There are two distinct finite-dimensionality statements. We prove that the three-relation quotient has a finite-dimensional quotient which is strictly above BMW and still detects X_3 . This is a finite-dimensional braid-group algebra answering the representation-theoretic part of the question. We do not prove that the universal three-relation quotient is finite-dimensional. This distinction is essential because the first twist relation already places the algebra in the setting of cubic Hecke quotients, where finite-dimensional universal quotients require additional structure; see [6, 7, 8, 9].

The source presentation contains a terminal index defect: its last displayed relation uses σ_n inside RB_n . We write $Z_n^\dagger$ for either natural well-typed repair: truncate at the last legal generator in RB_n , or interpret the final row in RB_{n+1} (equivalently in the direct-limit relation scheme). Every calculation below works separately under both interpretations.

2. BIGELOW'S ELEMENTS AND THE BURAU WITNESS

Bigelow uses

$$\sigma_{i_1 \dots i_r} = \sigma_{i_1} \cdots \sigma_{i_r}, \quad \bar{\sigma}_{i_1 \dots i_r} = \sigma_{i_1 \dots i_r}^{-1}.$$

Thus the bar denotes the inverse of the complete displayed word. Define

$$\begin{aligned} X_2 &= q\sigma_1^{-1} + 1 - q - \sigma_1, \\ X_k &= \left(q^{k-1} \bar{\sigma}_{(k-1) \dots 1} - \sigma_{1 \dots (k-1)} \right) X_{k-1} \quad (k \geq 3). \end{aligned}$$

Work over $\mathbb{Q}(q, u)$ and let $V = \mathbb{Q}(q, u)^n$ with basis $e_1, \dots, e_n$. Send σ_i to the unreduced Burau matrix B_i with nonidentity block

$$B_i|_{i, i+1} = \begin{pmatrix} 1-u & u \\ 1 & 0 \end{pmatrix}, \quad B_i^{-1}|_{i, i+1} = \begin{pmatrix} 0 & 1 \\ u^{-1} & 1-u^{-1} \end{pmatrix}.$$

For $1 \leq r < n$, put

$$a_r = u^{1-r} e_r - u^{-r} e_{r+1}, \quad b^T = -e_1^T + e_2^T, \quad E_r = a_r b^T.$$

Every E_r is nonzero of rank one.

Theorem 2.1 (Burau formula). *The unreduced Burau representation factors through each repaired presentation $Z_n^\dagger$, and for $2 \leq k \leq n$,*

$$\rho(X_k) = \left(\prod_{j=1}^{k-1} (q^j - u) \right) E_{k-1}.$$

In particular, $X_3 \neq 0$ over $\mathbb{Q}(q, u)$.

Proof. Direct calculation gives $\rho(X_2) = (q - u)E_1$. The local identities

$$\begin{aligned} (B_1 \cdots B_r) a_{r-1} &= u a_r, & (B_r \cdots B_1)^{-1} a_{r-1} &= a_r, \\ (B_2 \cdots B_{r+1}) a_r &= u a_{r+1}, & (B_{r+1} \cdots B_2)^{-1} a_r &= a_{r+1} \end{aligned}$$

follow from the two-by-two blocks. The first row of Bigelow's relations is a direct three-by-three calculation, and every later row follows from the shifted pair of identities. The same calculation handles the terminal row in the direct-limit repair. Induction in the defining recursion gives the formula. At $k = 3$, neither $q - u$ nor $q^2 - u$ vanishes in $\mathbb{Q}(q, u)$, and $E_2 \neq 0$. $\square$

3. TWIST EIGENRELATIONS

The first relation is immediate from $B_1 a_1 = -u a_1$.

Proposition 3.1. *In the generic Burau image,*

$$\rho(\sigma_1 X_2) = -u \rho(X_2).$$

For the higher relation, set

$$c_k = B_1 B_2 \cdots B_{k-1}, \quad W_k = \text{span}_{\mathbb{Q}(q,u)} \{a_1, \dots, a_{k-1}\}.$$

Lemma 3.2 (Full twist on the transported direction). *On W_k one has $c_k^k = u^k I$. Consequently,*

$$\rho\left((\sigma_1 \cdots \sigma_{k-1})^k X_k\right) = u^k \rho(X_k).$$

Proof. The Burau blocks give

$$c_k a_r = u a_{r+1} \quad (1 \leq r < k-1), \quad c_k a_{k-1} = -u(a_1 + \cdots + a_{k-1}).$$

Thus $T = u^{-1} c_k|_{W_k}$ is the companion operator of $1 + x + \cdots + x^{k-1}$. Hence

$$(I + T + \cdots + T^{k-1}) = 0, \quad T^k - I = (T - I)(I + T + \cdots + T^{k-1}) = 0.$$

Therefore $c_k^k = u^k I$ on W_k . Theorem 2.1 writes $\rho(X_k)$ as a scalar multiple of $a_{k-1} b^T$, so left multiplication by c_k^k multiplies it by u^k . $\square$

Corollary 3.3 (The twist requested for X_3).

$$\boxed{\rho((\sigma_1 \sigma_2)^3 X_3) = u^3 \rho(X_3)}.$$

Thus the two Burau twist scalars satisfy $\beta = -\alpha^3$ when α is the scalar for $\sigma_1 X_2$.

For comparison, $(B_1 B_2)^3$ is not scalar on the entire unreduced Burau module. The assertion is that it acts by u^3 on the rank-one left direction of $\rho(X_3)$, which is exactly what the zipper relation requires.

 4. THE SPECIALIZATION WITH $X_4 = 0$

Now specialize $u = q^3$ and work over $\mathbb{Q}(q)$. Denote the resulting Burau map by ρ_q . Theorem 2.1 yields

$$\begin{aligned} \rho_q(X_3) &= (q - q^3)(q^2 - q^3)E_2|_{u=q^3} \neq 0, \\ \rho_q(X_4) &= (q - q^3)(q^2 - q^3)(q^3 - q^3)E_3|_{u=q^3} = 0. \end{aligned}$$

The same recursion gives $\rho_q(X_k) = 0$ for every $k \geq 4$. Proposition 3.1 and Corollary 3.3 become

$$\boxed{\rho_q(\sigma_1 X_2) = -q^3 \rho_q(X_2), \quad \rho_q((\sigma_1 \sigma_2)^3 X_3) = q^9 \rho_q(X_3)}.$$

For $n \geq 4$, define the universal three-relation quotient

$$\mathcal{Q}_n = \mathbb{Q}(q) Z_n^\dagger / \langle \sigma_1 X_2 + q^3 X_2, ((\sigma_1 \sigma_2)^3 - q^9) X_3, X_4 \rangle.$$

Theorem 4.1 (Twist-compatible nonvanishing). *The representation ρ_q factors through $\mathcal{Q}_n$, and the image of X_3 has rank one. In particular, $X_3 \neq 0$ in $\mathcal{Q}_n$.*

Proof. All three added relations have zero image by the calculations above, while $\rho_q(X_3) \neq 0$. $\square$

The first twist relation can also be written as a cubic relation. If $s = \sigma_1$ and $sX_2 = \alpha X_2$, then multiplying by s and using the definition of X_2 gives

$$(s - 1)(s + q)(s - \alpha) = 0.$$

All standard braid generators are conjugate, so the corresponding relation holds for every σ_i . In $\mathcal{Q}_n$, where $\alpha = -q^3$, this is

$$\boxed{(\sigma_i - 1)(\sigma_i + q)(\sigma_i + q^3) = 0.}$$

This explains why the universal finite-dimensionality question belongs near the theory of cubic Hecke quotients.

5. A FINITE-DIMENSIONAL ALGEBRA STRICTLY ABOVE BMW

By Bigelow's quotient description, imposing $X_3 = 0$ and $\sigma_1 X_2 = tX_2$ on Z_n gives the BMW algebra. After the base change $t = -q^3$, the BMW quotient satisfies every defining relation of $\mathcal{Q}_n$: the first by definition, and the second twist and $X_4 = 0$ because $X_3 = 0$. There is therefore a surjection

$$\pi : \mathcal{Q}_n \twoheadrightarrow \text{BMW}_n(q, -q^3).$$

Let $A_n = \text{im}(\rho_q) \subseteq M_n(\mathbb{Q}(q))$ and define the joint image

$$D_n = \text{im}(\rho_q, \pi) \subseteq A_n \times \text{BMW}_n(q, -q^3).$$

Theorem 5.1 (Finite-dimensional extension quotient). *The algebra D_n is a finite-dimensional quotient of the braid group algebra $\mathbb{Q}(q)B_n$. Projection onto the second factor gives a surjection*

$$D_n \twoheadrightarrow \text{BMW}_n(q, -q^3)$$

with nonzero kernel. In particular,

$$\dim_{\mathbb{Q}(q)} \text{BMW}_n(q, -q^3) < \dim_{\mathbb{Q}(q)} D_n \leq \dim_{\mathbb{Q}(q)} \text{BMW}_n(q, -q^3) + n^2.$$

Proof. The Burau image A_n has dimension at most n^2 , and the BMW algebra is finite-dimensional by [4, 5]; hence the subalgebra D_n of their direct product is finite-dimensional. The second projection is surjective because π is. The element X_3 maps to

$$(\rho_q(X_3), \pi(X_3)) = (\rho_q(X_3), 0) \neq 0,$$

so the projection kernel is nonzero. The dimension inequalities follow. $\square$

Remark 5.2 (What remains open). Theorem 5.1 constructs a finite-dimensional braid-group algebra strictly above BMW and satisfying the requested twist and X_4 relations. It does not identify D_n with the universal quotient $\mathcal{Q}_n$ and does not prove $\mathcal{Q}_n$ finite-dimensional. Establishing a finite normal form for $\mathcal{Q}_n$, or comparing it with the diagrammatic zipper algebras reported in [3] and the current construction announced in [2], requires the complete later presentation.

6. EXACT VERIFICATION

The accompanying script `verify_twist_extension.py` reconstructs the Burau matrices from their two-by-two blocks. For $4 \leq n \leq 8$ it checks:

- (i) every well-typed displayed relation of $Z_n^\dagger$;
- (ii) the rank-one closed formula for every available X_k ;
- (iii) $\sigma_1 X_2 = -uX_2$ in the image;
- (iv) $(\sigma_1 \cdots \sigma_{k-1})^k X_k = u^k X_k$ in the image;
- (v) after $u = q^3$, both requested twist scalars, nonvanishing of X_3 , and vanishing of all X_k for $k \geq 4$.

The output is stored in `twist-extension-verification.json`. These finite symbolic checks are transcription safeguards; the proofs for all n are the block calculations above.

A second implementation, `verify_twist_extension_modular.py`, uses only pure Python finite-field matrices and no symbolic-algebra library. It checks the same defining and twist relations for $4 \leq n \leq 8$ over three prime fields, both at generic sample parameters and at $u = q^3$. Its output is `twist-extension-modular-verification.json`.

7. PRIORITY AND SCOPE

The 2004 BIRS workshop report states that Bigelow had described finite-dimensional zipper algebras properly extending BMW, but it does not provide the proof or the precise presentation used here. Bigelow's 2025 AMS abstract announces a large finite-dimensional algebra and a normal-form algorithm in joint work with Choomno Moos, again without publishing the defining relations. The present note gives explicit Burau twist scalars and the specialization $u = q^3$ for the printed 2005 relation scheme. No claim of worldwide priority is made before specialist comparison with Bigelow–Moos's current presentation and a complete literature review.

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